

EXP 2. SIMPLE LINEAR REGRESSION USING LEAST SQUARES METHOD

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In [1]: x = list(map(int, input("Enter x values: ").split()))
y = list(map(int, input("Enter y values: ").split()))

n = len(x)

x_bar = sum(x) / n
y_bar = sum(y) / n

cov_xy = sum((x[i] - x_bar) * (y[i] - y_bar) for i in range(n)) / n
var_x = sum((xi - x_bar) ** 2 for xi in x) / n

b = cov_xy / var_x
a = y_bar - b * x_bar

print(f"Regression equation: y = {a:.2f} + {b:.2f}x")
```

Regression equation: y = 0.00 + 2.00x